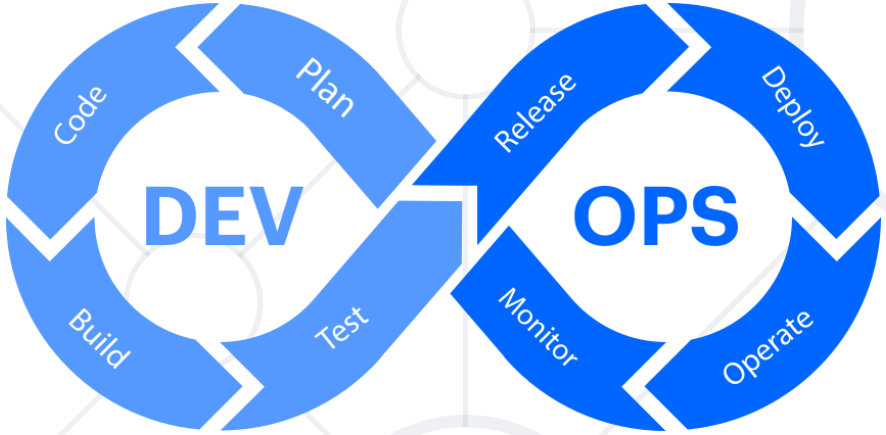
Security Integration

From DevOps to DevSecOps



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Have a Question?

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#DevOps-CI-CD

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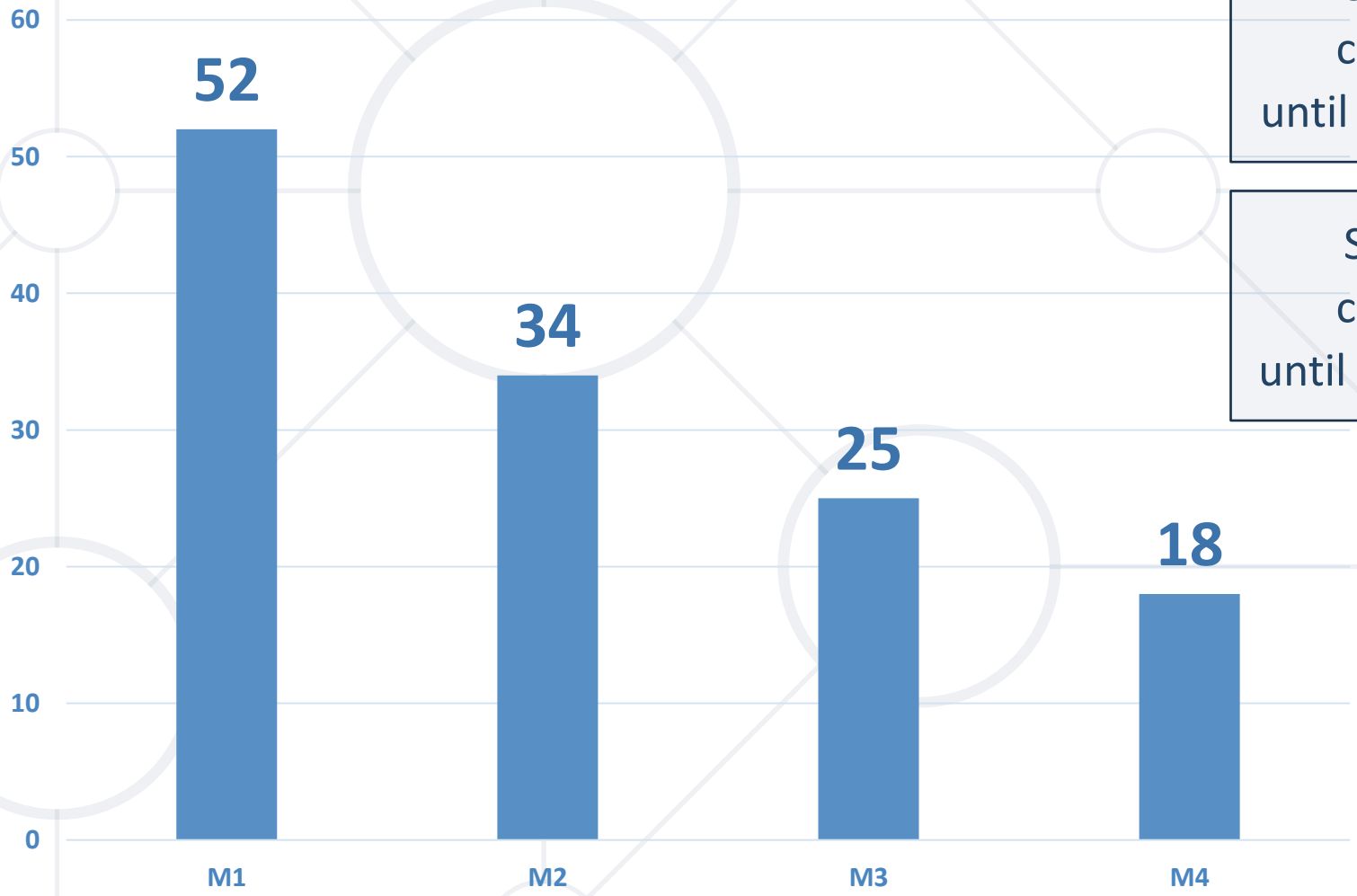
cicdmonitoringjanuary2026

Homework Progress



Solutions for M4
can be submitted
until 23:59 on 18.02.2026

Solutions for M5
can be submitted
until 23:59 on 25.02.2026

Previous Module (M4)

Quick Overview

What We Covered

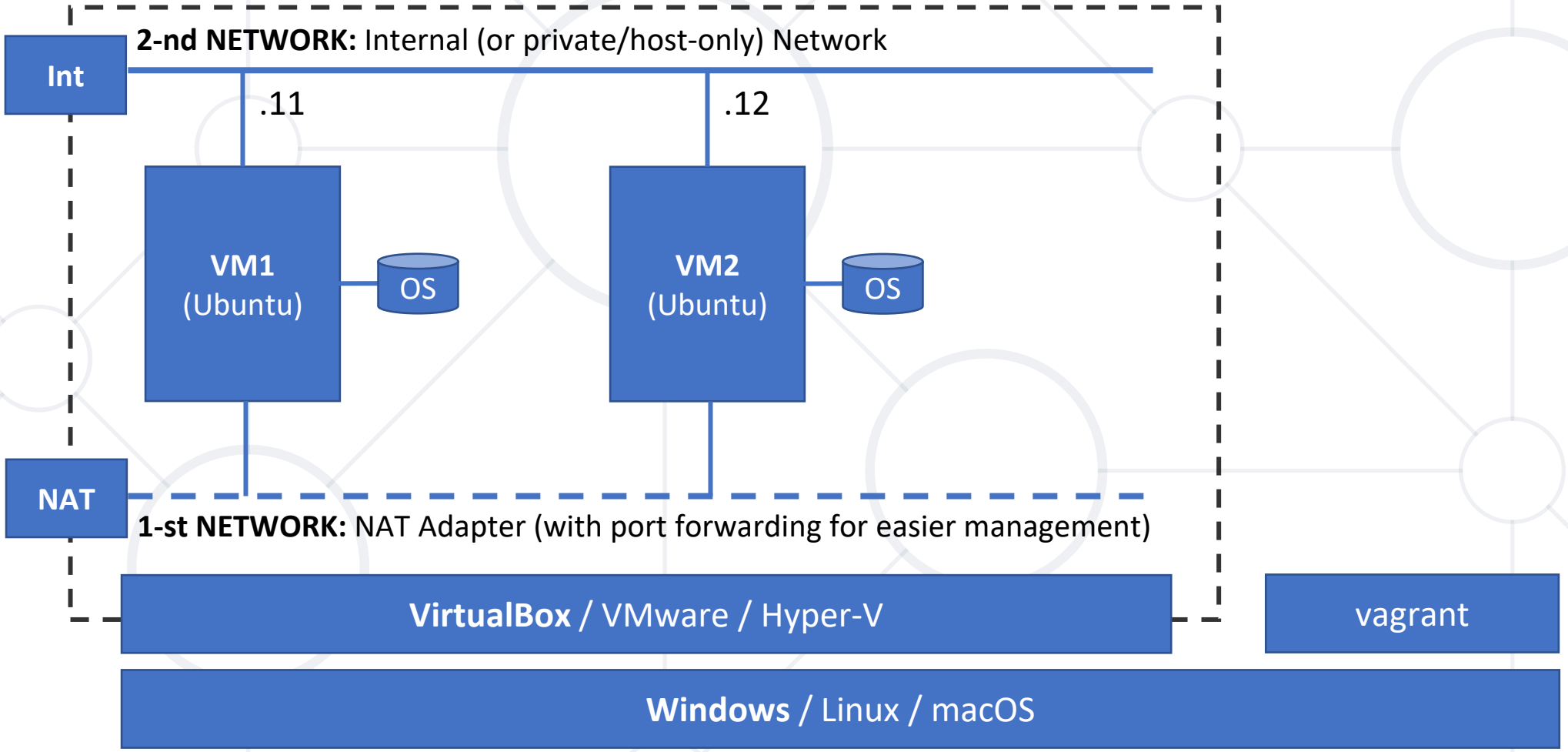
1. Packaging and Templating
2. Deployment and Organizational Strategies
3. Enable Automated Deployment

This Module (M5)

Topics and Infrastructure

Table of Contents

Lab Infrastructure



Shifting Left

From DevOps to DevSecOps

From DevOps to DevSecOps

- **The Shift Left Principle**
  - In traditional DevOps, security happened at the end (the Gate)
  - In DevSecOps, security starts at the first line of code
- **The Cost of Delay**
  - Fixing a bug in production is way more expensive than fixing it during development
- **Integration**
  - Security tools must be automated, fast, and fail the build if high-risk issues are found

The First Line of Defense

- **SAST (Static Application Security Testing)**
  - Scanning source code for **bad habits** (e.g., SQL injection, hardcoded creds)
  - **Linting** is a basic form of SAST
- **SCA (Software Composition Analysis)**
  - Scanning **package.json** or **requirements.txt**, or container images
  - Most modern apps are 90% third-party libraries

Our Approach and Tools

- Approaches
  - Decouple application components
  - Further parametrization
  - Initial secrets management
- Tools
  - Bandit
  - GoLangCI-Lint
  - HadoLint

Practice: Change Our Approach

Start implementing some of the practices ☺

Docker Compose

Linting and Secret Leaks

Secrets Leakage

- **The Pipeline Step**
  - Adding a stage that fails if it finds a string that looks like an **API key** or a **Private Key**
- **Best Practice**
  - Using **.gitignore** and **pre-commit hooks** to stop the leak before the git push

Our Approach and Tools

- Approaches
  - Reinvent Docker Compose
  - Further parametrization
  - Secrets management
- Tools
  - DCLint
  - GitLeaks

Practice: Linting and Secret Leaks

Let’s Move Forward

Kubernetes & Helm

Secrets Management and Hardening

Image Scanning

- **The Problem**
  - A **clean Python app** sitting on a **dirty Alpine** or **Debian base image** with **50 known vulnerabilities** (CVEs)
- **The Solution**
  - Scanning the image layer-by-layer after the build but ***before*** the push to the registry
- **The Tool**
  - **Trivy**, **Grype**, etc.

Infrastructure as Code Scanning

- **The Concept**
  - Helm charts and K8s manifests can be insecure (e.g., running as root, privileged mode, no resource limits, etc.)
- **Policy as Code**
  - Using tools like **Kube-score** or **Checkov** to analyze YAML files

Secret Management vs. ConfigMaps

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- **The Distinction**
  - **ConfigMaps** are for non-sensitive data
  - **Secrets** are for passwords, tokens, and certificates
- **Base64 is NOT Encryption**
  - Remember that K8s Secrets are just encoded, not encrypted by default
- **Approach**
  - Using Gitea/Jenkins **Secret Variables** to inject values into the pipeline at runtime

Sealed Secrets and External Vaults

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- **The Problem**
  - We want to store everything in Git, but we can't store secrets there
- **A Solution**
  - **Sealed Secrets** (Bitnami) - encrypt the secret with a public key; only the K8s cluster can decrypt it
- **A Pro Level Solution**
  - **HashiCorp Vault** - A dedicated server for managing secrets with automatic rotation

Network Policies

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- **Default Behavior**
  - In K8s, every pod can talk to every other pod
- **Zero Trust**
  - Implementing **NetworkPolicies** to ensure, for example, only the Frontend can talk to the API, and only the API can talk to the Database
- **The Blast Radius**
  - If the Frontend is hacked, the hacker cannot move to the Database

Our Approach and Tools

- Approaches
  - Address changes in Kubernetes manifests and Helm charts
  - Secrets management
  - Further hardening
- Tools
  - KubeConform
  - Kube-linter
  - Trivy

Practice: Manifests and Charts

Let’s Move Even Further

Questions?

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